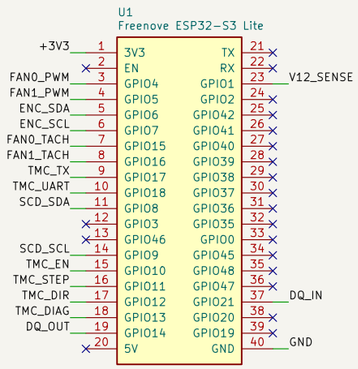
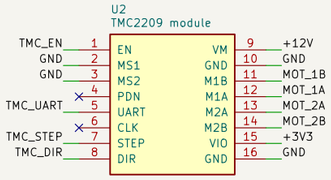
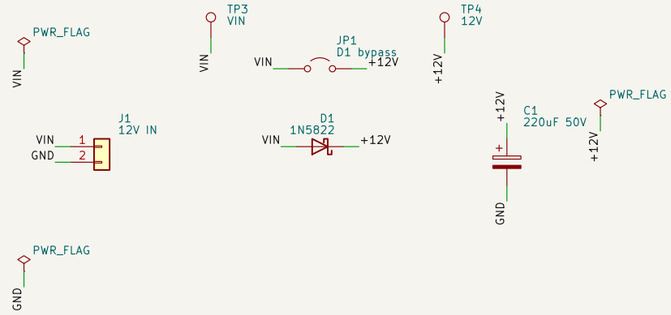
ESP32-S3 (Freenove Lite, FNK0099), soldered: USB toward the box's left wall



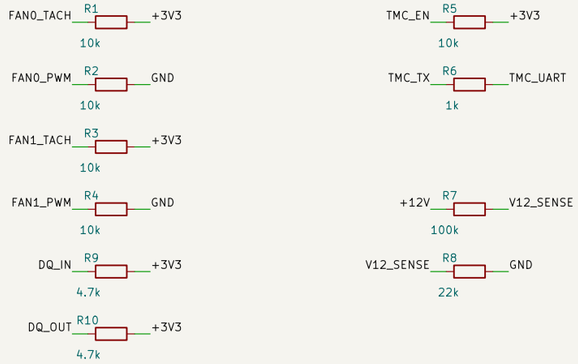
TMC2209 stepper driver (StepStick module), soldered



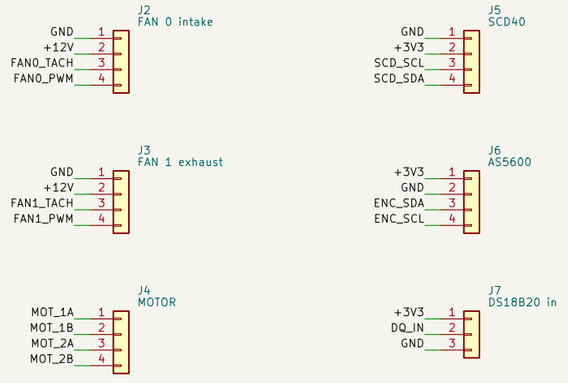
12 V input: D1 against reverse polarity; a jumper on JP1 bypasses it



Fan, temperature-probe and driver resistors; 12 V sense divider



Connectors (2.54 mm male headers)



Test pads



Mounting holes (M2)



Nets with the same label are connected. Pins as in firmware/baremetal/Pins.h.
U1 pins 1-20 are its left header (3V3 ... 5V) and 21-40 its right header (TX ... GND), top to bottom as on the board.
U2: MS1 = MS2 = GND (UART address 0); CLK and PDN free. DIAG is not on this module's pins: GPIO 13 goes to TP5.
R6 1k: single-wire UART, TX through 1k and RX straight onto PDN_UART. R5 holds the driver off (EN high) while the ESP32 boots.
R7/R8: 12 V reads 2.16 V on GPIO 1 (ADC1). R2/R4 keep the fans slow at boot; R1/R3 tach pull-ups; R9/R10 1-Wire pull-ups.
GPIO 38, 39, 40 stay free for the INMP441 microphone reserved in Pins.h (no room for its header on this board).
AS5600: tie its DIR pin to GND at the module. C1 sits next to U2's VM pin.

Generated by tools/pcb_schematic.py from tools/pcb_parts.py
github.com/mmassico/window_air_vent

Sheet: /
File: window_vent_ctrl.kicad_sch

Title: Window vent controller

Size: A3	Date: 2026-10-24	**Rev: 1.0**
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